

Samsung's Trade-In & Recycling Programs

A case study in e-waste management and the circular economy

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1. Executive Summary

Smartphones and other electronics are advancing faster than the systems that manage them once discarded. The world generated **62 million tonnes** of e-waste in 2022, only **22.3%** was formally collected and recycled, and volumes are projected to reach **82 million tonnes by 2030**. Yet a discarded device is not merely waste: the e-waste of 2022 alone contained about **31 billion kg of metals** including copper, aluminium, gold and cobalt.

This report examines **Samsung's Trade-In and Recycling Programs** as a case of a large manufacturer taking responsibility for its products beyond the point of sale. Samsung has collected e-waste since **2009**, operates collection systems in **100+ countries (2025)**, and reports a cumulative **7.54 million tonnes** collected by 2025, against targets of **10 million tonnes by 2030** and **25 million tonnes by 2050**.

The programme routes each returned device by condition: to **refurbishment** (Certified Re-Newed devices with a one-year warranty) or to **recycling** (dismantling and material recovery), with a closed-loop battery pathway that recovers cobalt for new batteries. In India the programme operates through Samsung Care for Clean India, inside the regulatory framework of the E-Waste (Management) Rules, 2022, which is built on Extended Producer Responsibility (EPR).

Key finding: cumulative collection is growing by roughly 0.6 million tonnes a year, which is encouraging, but collection volume alone does not prove circularity. The case data do not report how much of the collected material is reused, refurbished or recovered as materials, and the figures are Samsung-reported. Our recommendations therefore focus on outcome-based reporting, higher consumer participation, stronger data-privacy assurance, prioritising reuse over recycling, and integrating informal collectors in India.

2. Background

2.1 The global e-waste problem

Every smartphone poses a decision at end of life: it can sit forgotten in a drawer, end up in landfill or the informal sector, or enter a controlled circular system where it is returned, reused and recovered. The first two options waste remaining value and create environmental risk.

Table 1: Global e-waste context

Indicator	Value	Note
E-waste generated, 2022	62 million tonnes	Global generation
Formally collected and recycled	22.3%	Most e-waste is not formally recycled
Projected e-waste by 2030	82 million tonnes	Rising with technology turnover
Metals contained in 2022 e-waste	≈31 billion kg	Copper, aluminium, gold, cobalt and others

Source: *Global E-waste Monitor 2024 (ITU and UNITAR)*, as cited in the case slides.

2.2 A smartphone is not "just waste"

A phone contains copper, aluminium, gold and cobalt, along with plastics and glass. When these are not recovered, society loses valuable resources. When devices are processed unsafely, the consequences include hazardous substances, soil and water contamination, and air pollution from unsafe processing.

2.3 From linear to circular

The traditional linear model follows **extract → manufacture → buy → use → discard**. A circular model follows **design → use → trade-in → repair/refurbish → reuse, recover, remanufacture**, keeping products and materials at their highest possible value for as long as possible. Samsung's programmes are a practical example of shifting from "take–make–dispose" to "keep–recover–reuse".

2.4 Why Samsung?

- **Global electronics ecosystem:** phones, tablets, TVs, monitors and appliances.
- **Global collection network:** collection systems in 100+ countries (2025).
- **Long-term effort:** global e-waste collection since 2009.
- **Multiple strategies:** trade-in, repair, refurbishment, recycling and material recovery, offering a useful lens on how a large manufacturer manages products beyond point of sale.

3. Strategy / Initiative

3.1 Operating model

Not every returned device follows the same pathway. After a consumer returns a device through trade-in, its condition is assessed and it is routed to one of two paths:

Table 2: Operating model after condition assessment

Pathway	Steps	Outcome
Refurbishment path	Refurbishment → Certified Re-Newed → New user	Device gets a second life; product life is extended
Recycling path	Recycling → Dismantling → Material recovery	Components and materials return to manufacturing

3.2 Pillar 1: Trade-in and value preservation

Trade-in turns an upgrade into a return. The consumer selects the old device, describes its model and condition, receives an estimated value, purchases a new device, returns the old one, and the device is routed to reuse or recycling. In India, eligible phones, tablets and smartwatches from multiple manufacturers qualify.

The programme reflects a hierarchy of options, where more of the original value is preserved toward the top: **keep using → repair → refurbish/resell → recover components → recycle materials → dispose**. Refurbishment is not universally best, since condition, safety and market context still matter. Importantly, **trade-in is not recycling itself; it is the return mechanism into a controlled circular system**.

3.3 Pillar 2: Recycling

Recycling follows a seven-stage chain: **collection → sorting → dismantling → pre-processing → material recovery → processing → manufacturing**. Under Samsung Care for Clean India, the categories collected include smartphones, tablets, TVs, laptops, monitors, refrigerators, washing machines, air conditioners, microwave ovens and dishwashers. Recycling is not the end of the story: the goal is resource recovery.

3.4 Pillar 3: Refurbishment and closed-loop materials

Refurbished devices receive inspection, cleaning, refurbishment, testing, a certified battery, data reset and software updates, and carry a one-year warranty. The Galaxy S26 Certified Re-Newed is offered at up to about 85% of launch price.

The battery loop closes material flows: **used phone → battery collection → safe discharge → shredding → cobalt recovery → new material → new battery**. The Galaxy S24 was the first to incorporate recycled cobalt, and the S25 added advanced recovery from used batteries.

3.5 India's policy context

Samsung operates within a wider regulated system and does not manage India's entire e-waste system on its own. The **E-Waste (Management) Rules, 2022**, effective **1 April 2023**, place obligations on producers under **Extended Producer Responsibility (EPR)**. The chain runs: government regulation → producer responsibility →

Samsung/producer → collection system → authorised recyclers → material recovery, with consumer participation feeding the whole system.

4. Impact & Results

Samsung's reported cumulative collection has grown steadily. Table 3 combines the reported figures with calculations (marked "derived") that show progress against Samsung's stated targets.

Table 3: Samsung cumulative e-waste collected, 2009 to 2025

Year	Cumulative collected (M tonnes)	Increase on prior year (derived)	Progress to 2030 target of 10 M t (derived)	Progress to 2050 target of 25 M t (derived)
2023	6.30	n/a	63.0%	25.2%
2024	6.91	+0.61	69.1%	27.6%
2025	7.54	+0.63	75.4%	30.2%

Source: Samsung Electronics Sustainability (Samsung-reported figures); derived columns calculated by the authors.

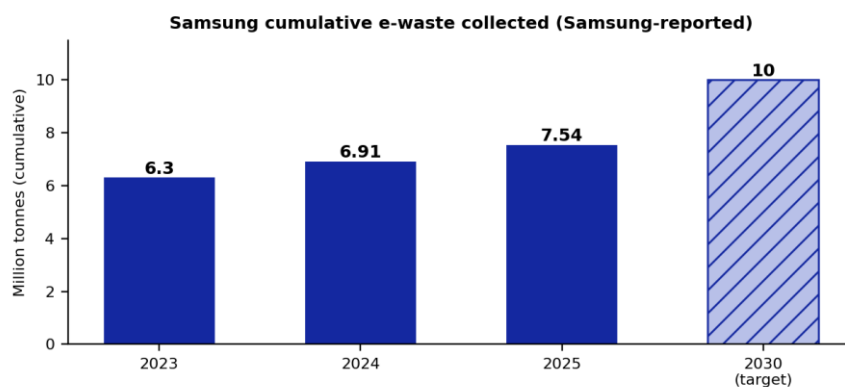


Figure 1: Cumulative e-waste collected versus the 2030 target.

Table 4: Programme reach and outcomes

Dimension	Result reported in the case
Collection reach	Collection systems in 100+ countries (2025); global collection since 2009
Cumulative collection	7.54 million tonnes (2009 to 2025)
Targets	10 million tonnes by 2030; 25 million tonnes by 2050
Refurbishment	Certified Re-Newed Galaxy S26 at up to about 85% of launch price, with one-year warranty
Closed-loop materials	Recycled cobalt first used in Galaxy S24; advanced battery recovery added in S25
India collection categories	10 product categories under Care for Clean India, from smartphones to dishwashers

Interpretation

- **Steady growth:** cumulative collection rose by about 0.61 M t in 2024 and 0.63 M t in 2025. Continuing at that pace would take Samsung past 10 M t before 2030, since only about 0.49 M t a year is needed for 2026 to 2030.
- **The 2050 target is more demanding:** reaching 25 M t needs an average of about 0.70 M t a year from 2026, above the current pace, so annual collection must keep rising.

- **Scale in context:** Samsung's 16-year cumulative total (7.54 M t) equals only about 12% of the 62 M t the world generated in the single year 2022.
- **Limits of the data:** collection volume is not the same as circularity. The case does not report reuse or refurbishment rates, material recovery rates or recycled-content share, and the collection figures are Samsung-reported and not independently verified. A complete assessment weighs reuse, refurbishment, material recovery and environmental outcomes.

5. SWOT Analysis

The SWOT below evaluates Samsung's trade-in and recycling model. Strengths and weaknesses are internal to the programme; opportunities and threats arise from the external environment.

Strengths • Long-running global effort (since 2009) with reach in 100+ countries • Integrated pathways: trade-in, refurbishment, recycling and material recovery • Closed-loop battery cobalt recovery from S24 and S25 onwards • Certified Re-Newed offer with one-year warranty extends product life • Cumulative 7.54 M t collected with clear 2030 and 2050 targets	Weaknesses • Trade-in does not guarantee reuse; not all devices are refurbished • Collection volume is emphasised over reuse and recovery outcomes • Figures are self-reported, with limited detail on recovery rates • Product complexity makes material recovery difficult
Opportunities • Growing e-waste (62 M t to 82 M t by 2030) expands the recoverable resource base • India's EPR rules (2022) create a structured framework for collection and recycling • Rising consumer interest in certified refurbished devices • Recovered metals can reduce dependence on virgin mining	Threats • Consumer reluctance to return old devices • Data privacy concerns about personal information on devices • Uneven infrastructure and recycling capability across countries • Informal recycling channels competing for used devices • Regulatory and compliance risk if targets or audits tighten

6. Comparison with India

This section compares Samsung's global programme with India's national e-waste system, into which Samsung India's activities fit. India's regulatory position is drawn from the E-Waste (Management) Rules, 2022 and Central Pollution Control Board (CPCB) material. The two are different kinds of system, so the figures are not directly comparable and the table is meant to highlight structural differences.

Table 5: Samsung's programme versus India's national system

Dimension	Samsung programme (global, with India activity)	India's national system
Basis	Corporate programme running since 2009 across 100+ countries	Statutory: E-Waste (Management) Rules, 2022, in force from 1 April 2023
Who is responsible	Manufacturer-led collection, refurbishment and recycling	Shared: producers, refurbishers and recyclers must register on the CPCB portal; only registered entities may collect and process e-waste
Targets	10 M t cumulative by 2030; 25 M t by 2050	Annual EPR targets rising from 60% (FY 2023-24 and 2024-25) to 70% (FY 2025-26 and 2026-27) to 80% (from FY 2027-28) of generation-based obligation
Scale indicator	7.54 M t cumulative collected worldwide (2025)	About 1.6 M t of e-waste generated in FY 2021-22 across 21 notified categories (CPCB), more than double FY 2017-18
Collection channels	Trade-in for phones, tablets and smartwatches; Care for Clean India takes 10 product categories	Producer take-back systems, authorised recyclers and refurbishers; EPR certificates link recycling to producer compliance
Informal sector	Controlled chain: sorting, dismantling and recovery through authorised partners	Significant informal collection and recycling using unscientific methods; the Rules aim to move informal players into the formal sector
Accountability	Self-reported sustainability figures	CPCB portal tracking, audits and environmental compensation for non-compliance

Sources: case slides; Samsung India (Care for Clean India); MoEFCC E-Waste (Management) Rules, 2022; CPCB FAQ; PIB release; IMPRI summary of CPCB data.

What the comparison shows

- **Complementary roles:** the Rules set the obligations and enforcement, while a producer like Samsung supplies the collection network, brand pull for trade-in and technical capability for refurbishment and recovery.
- **Voluntary reach versus compulsory floor:** Samsung's global targets are ambitions, while India's EPR targets are legally binding percentages of generation, backed by environmental compensation.
- **The India-specific challenge is leakage to the informal sector.** A strong producer programme helps only if devices actually reach it, which depends on consumer participation and convenient, trusted collection.
- **Measurement differs:** India measures obligations against generation estimates and certificates, whereas Samsung reports cumulative tonnes collected. Neither by itself shows how much material is truly reused or recovered.

7. Recommendations

The recommendations follow from the weaknesses and threats identified above and from the case's central question, whether collection volume is enough to measure circularity.

For Samsung and other manufacturers

- **Report outcomes, not just tonnes.** Publish the share of collected devices that are reused, refurbished, recovered for components, and recycled to materials, plus recycled-content share in new products. Seek independent verification of reported figures.
- **Prioritise reuse before recycling.** Follow the value hierarchy (keep, repair, refurbish, recover, recycle) by expanding repair access and Certified Re-Newed volumes, and design products for easier disassembly to address recovery difficulty.
- **Raise consumer participation.** Offer simple pick-up or drop-off, clear trade-in valuations and incentives, so that fewer devices remain in drawers or move to the informal sector.
- **Build trust on data privacy.** Communicate and certify secure data erasure at collection and again at refurbishment, since concern about personal information is a major reason for reluctance to return devices.
- **Scale closed-loop materials.** Extend recycled cobalt beyond flagship models and add other recovered metals, reporting how much recovered material re-enters new products.

For policy makers in India

- **Strengthen enforcement and traceability** of EPR certificates and recycler activity, including audits, so that certificates reflect real material flows.
- **Integrate informal collectors** into formal channels through registration, training and linkages with authorised recyclers, rather than leaving them to compete with the formal system.
- **Recognise reuse in targets and reporting**, so that refurbishment and life extension are encouraged alongside tonnage recycled.

For consumers and institutions

- Return devices through authorised channels, erase personal data before handing over, and consider repair or refurbished purchases before buying new.

8. Conclusion

Samsung's trade-in and recycling programmes show how a manufacturer can manage products beyond the point of sale through three connected strategies: **extend** product life through repair, trade-in and refurbishment; **recover** components and materials when reuse is impractical; and **close the loop** by returning recovered materials, such as cobalt, to productive use.

With 7.54 million tonnes collected since 2009, a network spanning 100+ countries and ambitious 2030 and 2050 targets, the programme demonstrates real scale. In India, it operates within a regulated EPR framework that complements corporate effort. However, challenges remain: consumer reluctance to return devices, data privacy concerns, complex products that are hard to recover, uneven infrastructure across countries, and the fact that trade-in does not always mean reuse.

The answer to the case's central question, "Is collection volume enough to measure circularity?", is therefore **no**. A complete assessment weighs reuse, refurbishment, material recovery and environmental outcomes, not collection volume alone. As the case summarises, the goal is not simply to recycle yesterday's technology but to prevent today's technology from becoming tomorrow's waste.

9. References

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- [10] Case study slides: Samsung's Trade-In & Recycling Programs, Environmental Management, 23 September 2026 (primary source for case details and Samsung-reported figures).

10. One-Page Exam Revision Sheet: Samsung Trade-In & Recycling (Circular Economy)

Core idea: Samsung uses **trade-in as the return mechanism** into a controlled circular system (refurbish, recycle, recover). Goal: move from "take–make–dispose" to "keep–recover–reuse".

KEY NUMBERS

- Global e-waste 2022: **62 M t**; formally recycled **22.3%**; **82 M t** by 2030
- Metals in 2022 e-waste: **≈31 billion kg**
- Samsung cumulative: **6.30 (2023) → 6.91 (2024) → 7.54 M t (2025)**; growth ≈ 0.6 M t/yr
- Targets: **10 M t by 2030, 25 M t by 2050**; 100+ countries; since **2009**
- Re-Newed Galaxy S26: up to **~85%** of launch price, 1-yr warranty
- Progress: 75.4% of 2030 target, 30.2% of 2050 target (derived)

KEY TERMS

- **E-waste:** discarded electronics with hazardous substances and recoverable materials
- **Circular economy:** keep products/materials at highest value as long as possible
- **EPR:** producers' obligation to manage post-consumer waste
- **Closed loop:** recovered materials go back into new products (e.g. cobalt to batteries)
- **Trade-in $\neq$ recycling:** it is the return route; not all devices get refurbished

VALUE HIERARCHY (top = most value kept)

Keep using → Repair → Refurbish/Resell → Recover components → Recycle materials → Dispose

SAMSUNG'S 3 STRATEGIES

- **Extend:** repair, trade-in, refurbish
- **Recover:** components and materials when reuse impractical
- **Close the loop:** return recovered materials to use

OPERATING MODEL

New device → Use → Trade-in/Return → **Condition assessment** → (A) Refurbish → Certified Re-Newed → New user, or (B) Recycle → Dismantle → Material recovery

PROCESS CHAINS

- **Recycling (7):** Collection, Sorting, Dismantling, Pre-process, Material recovery, Processing, Manufacturing
- **Battery loop:** Used phone → Collect → Safe discharge → Shred → Cobalt recovery → New material → New battery (S24 first recycled cobalt; S25 advanced recovery)
- **Trade-in (6):** Select device, describe condition, get estimate, buy new, return old, route to reuse/recycle

INDIA CONTEXT

- **E-Waste (Management) Rules, 2022**, effective **1 Apr 2023**, EPR-based
- Chain: Govt regulation → Producer responsibility → Producer → Collection → Authorised recyclers → Material recovery (+ consumer participation)
- EPR targets: 60% → 70% → 80% of obligation; CPCB portal, certificates, environmental compensation
- Care for Clean India: 10 categories (phones to dishwashers)
- Samsung does not run India's whole system alone

SWOT IN BRIEF

- **S:** global reach, integrated pathways, closed-loop cobalt, targets
- **W:** trade-in $\neq$ reuse; volume over outcomes; self-reported; complex products
- **O:** rising e-waste, India EPR framework, demand for refurbished, recovered metals
- **T:** consumer reluctance, data privacy, uneven infrastructure, informal sector

CRITICAL EVALUATION (5 challenges)

Consumer participation · Data privacy · Product complexity · Infrastructure varies · Trade-in $\neq$ reuse

LIKELY EXAM QUESTIONS

- **Q: Is collection volume enough to measure circularity?**
No. Also weigh reuse, refurbishment, material recovery and environmental outcomes.
- Explain trade-in versus recycling; draw the operating model.
- Linear vs circular; role of EPR; SWOT of Samsung's model.

ONE-LINE TAKEAWAY

"Not simply to recycle yesterday's technology, but to prevent today's technology from becoming tomorrow's waste."